STUDENT MARKETPLACE APPLICATION

PROBLEM STATEMENT

University students often need to buy and sell items such as textbooks, clothes, electronics, stationery, and other useful products. However, finding affordable items or reaching other students who want to buy them can be difficult. Some students advertise items through WhatsApp groups, social media platforms, or word of mouth. These methods can make it difficult to find products quickly, compare prices, contact sellers, and identify relevant listings. Students may also struggle to find a suitable place to advertise their products specifically to the university community.

The Student Marketplace is a proposed mobile application designed to help students buy and sell items conveniently within their university community. The system will allow students to register and log in, create product listings, upload product images, provide descriptions, set prices, search for available products, and view product details. Students who are interested in an item will be able to contact the seller using the contact information or communication method provided by the system. Sellers will be able to edit or remove their listings when necessary.

The system will also provide administrators with tools to manage users and monitor product listings to help reduce inappropriate or misleading advertisements. The application will organise products into categories, making it easier for students to find items of interest. By bringing buyers and sellers together on one platform, the Student Marketplace aims to save students time, improve access to affordable products, and provide a more organised way of trading within the university community.

The proposed system will initially focus on product advertising and connecting buyers with sellers. Unless additional features are approved, it will not process payments directly or provide delivery services. Buyers and sellers will arrange payment and collection themselves. The application will be developed with usability in mind so that students with different levels of technical experience can use it easily.

2. Users and their goals

There are TWO main users in the system.

Students and Administrator.

User Goals

Student Register, log in, browse products, search by name or category, view prices and descriptions, and contact sellers , upload images, set prices, edit listings, and remove sold or unavailable items.

Administrator Monitor listings, manage user accounts, remove inappropriate content, and help keep the marketplace organised.

3. System requirements

3.1 Functional requirements

Functional requirements describe what the system must do.

FR1 — Registration

The system shall allow students to create an account using their name, email address and password.

FR2 — Login

The system shall allow registered users to log in using their credentials.

FR3 Profile management

The system shall allow users to view and update their profile information.

FR4 Add products

The system shall allow sellers to create product listings containing a name, description, price, category and image.

FR5 Browse products

The system shall allow users to browse available products.

FR6 Search products

The system shall allow users to search for products by name or keyword.

FR7 Filter products

The system shall allow users to filter products by category and price.

FR8 View details

The system shall display product details, price and seller information.

FR9 Manage listings

The system shall allow sellers to edit or remove their own product listings.

FR10 Place orders

The system shall allow buyers to place orders for available products.

FR11 Manage orders

The system shall allow buyers and sellers to view their relevant orders and order statuses.

FR12 Administration

The system shall allow administrators to manage user accounts and remove inappropriate product listings.

3.2 Non-functional requirements

Non-functional requirements describe how well the system should work.

NFR01 Performance .

The system should display normal product listings within 3 seconds under typical operating conditions.

NFR02 Usability .

The system shall provide a simple, consistent interface that students can learn to use without extensive training.

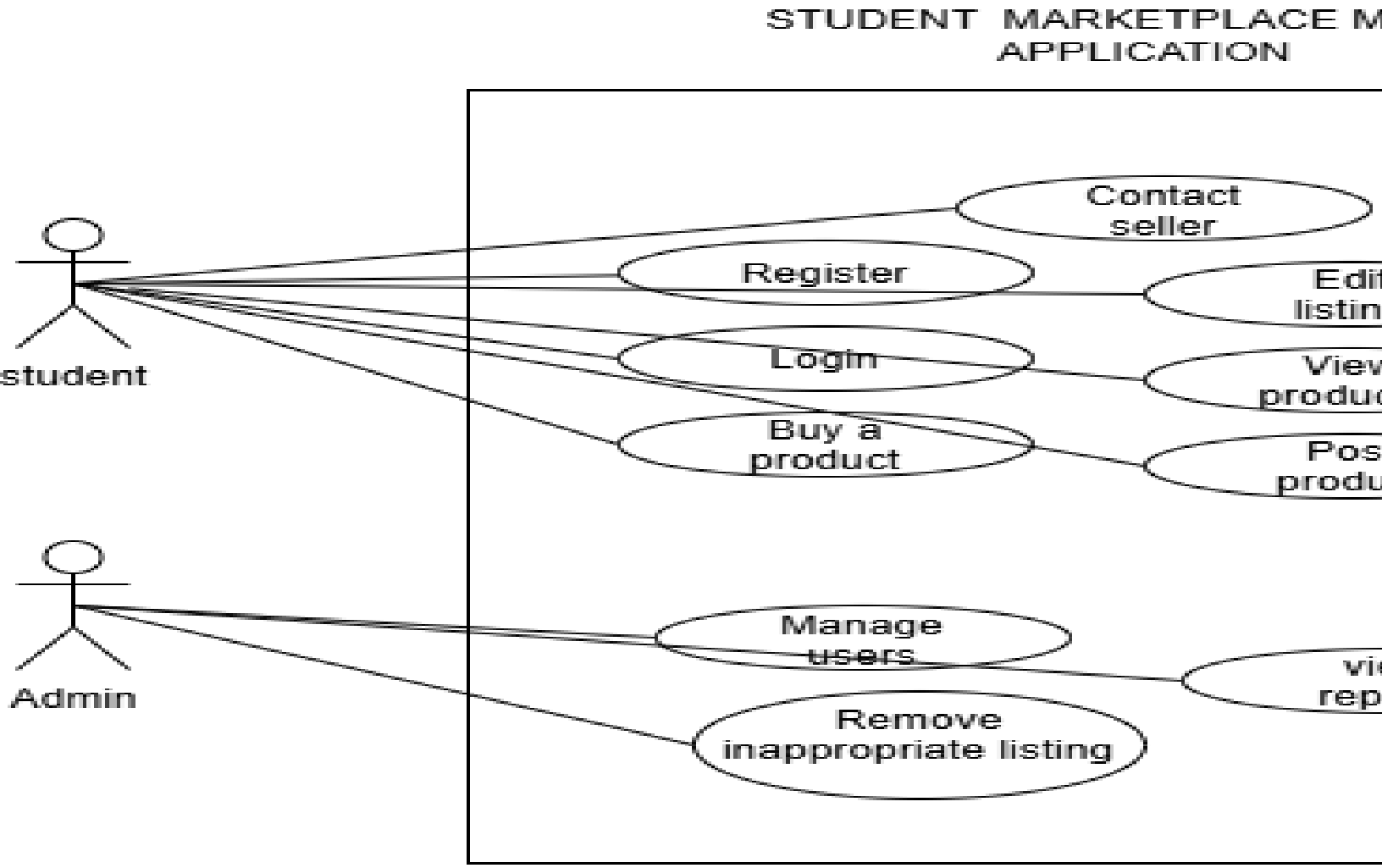
NFR03 Security.

The system shall store passwords securely using an appropriate password-hashing method and restrict protected actions to authorised users.

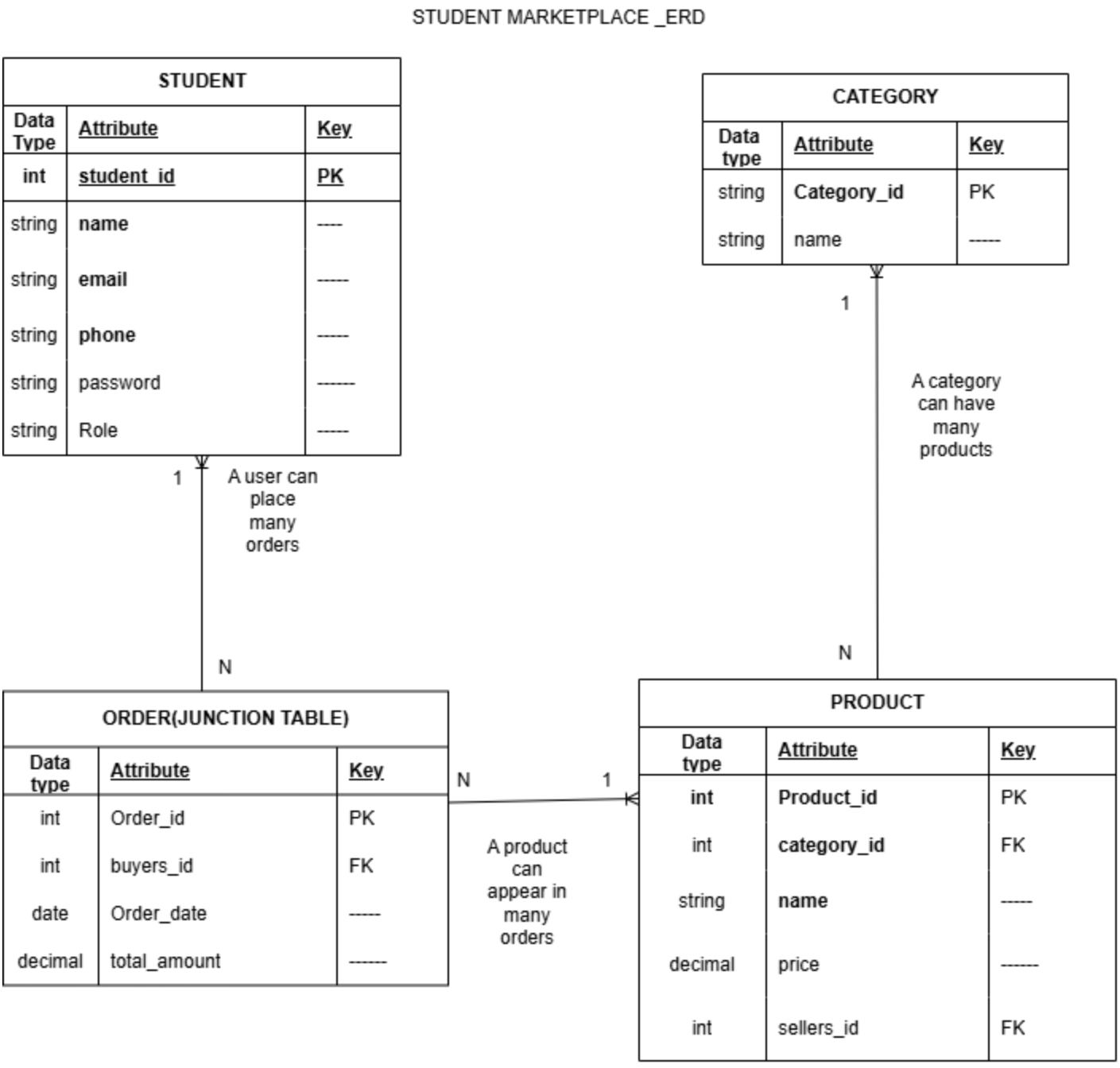
NFR04 Reliability.

The system shall validate submitted data and display understandable error messages when an operation fails.

3. Use case diagram.


STUDENT MARKETPLACE M
APPLICATION
Contact seller
Register
Edit listin
Login
View produc
Buy a product
Pos
produ
Manage users
vi
rep
Remove inappropriate listing
student
Admin

5.ERD



6. Data dictionary

A data dictionary explains every field in your database: its name, type, size, rules and meaning.

Users table

Field	Type and size	Constraints	Meaning
User_id	INT	PRIMARY KEY	Unique identifier
Name	String varchar(40)	NOT NULL	Users name
Phone	String varchar(20)	NULL	Contact number
Password	String varchar	NOT NULL	Secure password
Role	String varchar(20)	NOT NULL default student’	Student or admin

Category table

Field	Type and size	Constraints	Meaning
Category_id	INT	PK	Unique category ID
Name	string	NOT NULL	Category name

Product table

Field	Type and size	Constraints	Meaning
Product_id	INT	PK	Unique product ID

Category_id	INT	FK	References categorical.category_id
Name	STRING	NOT NULL	Product title
Price	DECIMAL	NOT NULL	Selling price
Sellers_id	INT	FK	References users.user_id

Order table

Field	**Type and size**	**Constraints**	**Meaning**
Order_id	INT	PK	Unique orderID
Buyers_id	INT	FK	References user.user_id
Order_date	DATE	NOT NULL	Date and time of the order
Total_amount	DECIMAL(12,5)	NOT NULL	Total order value

7. API endpoint table

An API endpoint is an address through which Flutter communicates with the FastAPI backend.

For example, when a student opens the product page, Flutter requests product data from FastAPI, and FastAPI retrieves the data from MySQL and returns it.

The following is a proposed API specification.

Method	**Path**	**Purpose**
POST	/api/auth/register	Create an account
POST	/api/auth/login	Authenticate a user
POST	/api/user/me	View the logged-users profile
GET	/api/categories	List product categories
GET	/api/products	Browse products;supports search and filter
GET	/api/products/{product_id}	View one product
POST	/api/products	Create a listing
PUT	/api/products/{product_id}	Update an owned listing
DELETE	/api/products/{product_id}	Remove an owned listing.
POST	/api/orders	Create an order
GET	/api/orders	View the logged-in users relevant orders

9:41

ISLAMIC CALL UNIVERSITY

Hey, 👋

Search your campus

LESS SPENDING. MORE LIVING.

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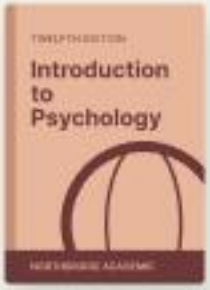
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Order summary

Textbook	shs.40,000
Campus pickup	Free

Total to pay **SHS.40,000**

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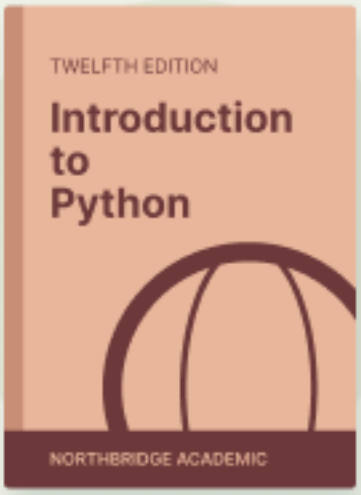
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The details



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STUDENT MARKETPLACE

Good finds.
Great campus.

Buy, sell and pass good things on.



Student | Administrator

University email

enter your email

Password

••••••••••

Forgot password?

Sign in →

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